Track #1: Crypto Factor Rebalancing Strategy

Artemis Analytics Quant Research / Trading Strategy Competition

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Network-Augmented Latent Factor Portfolio: A Comprehensive Research Summary

Prepared for: Artemis Analytics Quant Research / Trading Strategy Competition

Track: #1: Crypto Factor Rebalancing Strategy
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Codebase: `08_nalfp/` (NALFP v2)

Upstream Contributions:
* `04_factors/`
* `06_artemis_econometrics/`
* `07_hidden_factor_pricing/`

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1. Research Overview

This report summarises a multi-stage quantitative research effort to construct a weekly-rebalanced, long/short cryptocurrency factor strategy, submitted in response to **Track #1 (Crypto Factor Rebalancing Strategy)** of the Artemis Analytics Quant Competition. The competition challenges teams to "apply quantitative methods to real crypto markets," evaluating submissions on four equally weighted dimensions: Research Quality (30%), Signal/Edge Validity (30%), Critical Evaluation (20%), and Communication (20%). Crucially, the organisers state: *"We are not looking for the most profitable strategy — we are looking for the best thinking. A strategy with a negative Sharpe ratio that is correctly understood, honestly evaluated, and clearly presented will score better than a strategy with impressive backtested returns and no critical analysis."* This principle has guided the structure of the present report.

The project draws on four published or working papers in crypto asset pricing, implements each as a reproducible pipeline stage, and integrates their outputs into a unified portfolio we term the **Network-Augmented Latent Factor Portfolio (NALFP)**. The strategy addresses each Track #1 requirement as follows:

Competition Requirement	How NALFP Addresses It
Define a clear investment universe	Top ~80 crypto assets by lagged market cap, excluding stablecoins/wrapped/bridged tokens; median portfolio of 16 long + 16 short names per week
Identify one or more factors or signals	Nine economically-motivated factor portfolios + network-topology signals + adaptive IC-weighted blend
Specify rebalancing frequency and portfolio construction	Weekly rebalance; long top quintile / short bottom quintile; inverse-vol weighting; dollar-neutral; cluster cap, turnover budget, vol target
Backtest across a meaningful historical window and report performance	52-week sample, 24-week OOS walk-forward; returns, Sharpe, max drawdown, turnover, hit rate, bootstrap CI all reported
Critically evaluate findings	Negative OOS Sharpe analysed in depth; honest limitations section; regime-specific failure modes identified

1.1 Pipeline Architecture

The research progressed through six active stages, each building on the prior:

Stage	Directory	Purpose	Output
Data Collection	`01_Data_Collection/`	Weekly price, volume, on-chain activity panels	Clean parquet tables
Analysis	`03_analysis/`	EDA, data quality, universe profiling, cross-sectional diagnostics	Diagnostics

Stage	Directory	Purpose	Output
Factor Signals	`04_factors/`	Three novel factors (F, S, G) + RAAM v2 IC-weighted composite	Factor ranks and IC tables
BTC Direction	`05_btc_direction/`	Supervised BTC direction classifier	Standalone; not integrated into NALFP
Econometrics	`06_artemis _econometrics/`	Fama–MacBeth panel with characteristics + PCA controls + network features	Model predictions, IC/IR tables
Hidden Factor Pricing	`07_hidden_factor _pricing/`	Observed vs. latent-adjusted factor pricing (GX-lite)	Factor premia and survival table
NALFP	`08_nalfp/`	Integrated strategy: network + factor pricing + adaptive blend + portfolio construction	Final portfolio weights and backtest

1.2 Data Universe

The panel comprises approximately 86–113 crypto assets observed over 52 weeks (2025-05-12 to 2026-05-04), sourced from CoinGecko daily ticks, Artemis on-chain activity (DAU, fees, revenue), and DeFiLlama (TVL, stablecoin flows). Stablecoins, wrapped tokens, and bridged assets are excluded via `coingecko_coin_details` flags. All characteristics are lagged one week prior to entering predictive models.

1.3 Key Design Decision

NALFP v1 employed an Instrumented PCA (IPCA) framework (Kelly, Pruitt, Su 2019), where lagged characteristics instrument time-varying factor loadings. This approach was replaced in v2 with explicit factor portfolios plus the Giglio–Xiu (2021) three-pass procedure. The rationale, as documented in `RESEARCH_PLAN.md`, was that "v1 IPCA obscured *which* factors the strategy was actually exposed to and lacked per-factor economic commentary." The v2 architecture produces a transparent, per-factor risk-premium table directly comparable with prior literature.

Competition alignment. Track #1 of the Artemis Quant Competition requires: (1) a defined investment universe, (2) identified factors or signals, (3) a rebalancing methodology, (4) a backtest with standard performance metrics, and (5) a critical evaluation of limitations. This strategy addresses all five requirements. The judging criteria — Research Quality (30%), Signal/Edge Validity (30%), Critical Evaluation (20%), Communication (20%) — are addressed throughout this report, with particular emphasis on honest assessment of the negative OOS Sharpe (Section 5) and structural limitations (Appendix C).

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2. Core Trading Strategy

The NALFP strategy combines three pillars into a single weekly-rebalanced long/short portfolio.

2.1 Pillar 1 — Time-Varying Network Structure

Implementation: `08_nalfp/01_network_dynamics.py`

Each Monday, a rolling 12-week Spearman correlation matrix is computed across all assets with sufficient non-missing overlap (8 shared weeks). The correlation matrix is converted to a Mantegna distance matrix via $d_{ij} = \sqrt{2(1 - \rho_{ij})}$, from which a Minimum Spanning Tree (MST) is extracted. Louvain community detection (resolution = 1.0, seed = 0) is then applied to the MST (with edge weights transformed as $1/(1 + d)$ for similarity).

Three per-asset signals are emitted:

Signal	Construction	Intuition
`within_cluster_mom`	Rank-z-score of 4-week momentum within each Louvain cluster	Idiosyncratic momentum relative to the local community
`cross_cluster_rel`	Asset 4-week return minus the mean 4-week return of *other* clusters	Cross-narrative rotation
`cluster_id`	Integer community label	Used for portfolio-level diversification constraint

One market-level signal is also emitted:

Signal	Construction	Intuition
`network_entropy`	Shannon entropy $H = -\sum_k p_k \ln p_k$ of cluster-size distribution	Fragmentation proxy; high $H \to$ many small disconnected communities

The manifest (`01_network_manifest.json`) records: 41 clustered weeks, 7–11 clusters per week, mean entropy 2.175. All quantities are computed using only data through $t-1$ (strict no-lookahead).

2.2 Pillar 2 — Crypto Factor Zoo + Giglio–Xiu Pricing

Implementation: `08_nalfp/02_factor_pricing.py`

Nine economically-named long/short factor portfolios are constructed weekly:

Factor	Sort Variable	Direction	Leg Size	Source
RC	Value-weighted market return	—	Universe	Hartmann 2025
SMBC	`log_mcap`	Small-long, Big-short	30/30	Hartmann 2025; Fama–French 1993
MomC	`mom_4w`	Winners-long, Losers-short	30/30	Hartmann 2025; Liu–Tsyvinski 2022
VolC	`vol_4w`	Low-vol long, High-vol short	30/30	Frazzini–Pedersen 2014
TVLC	`tvl_to_mcap`	High TVL long, Low TVL short	30/30	Hartmann 2025
FunC	`F_yield`	High yield long, Low yield short	30/30	RAAM v2 stage 04
SupC	`S_supply`	Low emission long, High emission short	30/30	RAAM v2 stage 04
NetMom	`within_cluster_mom` (cluster-neutral)	Top-half long, Bottom-half short	50/50 within cluster	Liu & Tsyvinski 2018

Factor	Sort Variable	Direction	Leg Size	Source
NetRel	`cross_cluster_rel`	Outperformers long, Underperformers short	30/30	Liu & Tsyvinski 2018

Giglio–Xiu three-pass procedure (walk-forward, refit every 4 weeks on expanding window):

1. **Pass 1:** OLS time-series regression of each asset's weekly return on the 7 observed factors (RC, SMBC, MomC, VolC, FunC, NetMom, NetRel — TVLC and SupC are dropped due to sparse coverage: TVLC has < 30 weeks, SupC only 7 weeks).
2. **Pass 2:** PCA on the residual matrix; Bai–Ng (2002) IC_{p2} criterion selects **K_hidden = 3** latent factors. (Note: IC_{p2} is monotonically decreasing across $K \in \{0, 1, 2, 3\}$; the $K = 3$ cap is hit rather than a clear elbow, suggesting identification may improve with a longer sample.)
3. **Pass 3:** Refit β on the combined factor set [observed + hidden]; cross-sectional Fama–MacBeth estimation of risk premia $\hat{\lambda}$ with heteroskedasticity-robust (White) standard errors.

The output per asset per week is $E_{gx,it} = \beta_i' \hat{\lambda}$, the expected next-week return implied by the factor model.

2.3 Pillar 3 — Adaptive Regime Blend

Implementation: `08_nalfp/03_regime_detector.py`

Two cross-sectional signals are available each week: $E_{net,it}$ (within-cluster momentum z-score from Pillar 1) and $E_{gx,it}$ (Giglio–Xiu expected return from Pillar 2). The blended expected return is:

$$E_{final,it} = w_{net,t} \cdot E_{net,it} + (1 - w_{net,t}) \cdot E_{gx,it}$$

where $w_{net,t}$ is set by the **out-of-sample rolling 8-week mean Spearman IC** of each signal against next-week returns:

$$w_{net,t} = \frac{\max(0, \bar{IC}_{net,t-1})}{\max(0, \bar{IC}_{net,t-1}) + \max(0, \bar{IC}_{gx,t-1})}$$

If both rolling ICs are non-positive, the weight defaults to 0.5. This is the same IC-weighting rule validated in stage 04 (RAAM v2 composite) and is deliberately mechanical — no tuning parameters — so that a reviewer can verify it directly from the code.

The manifest (`03_regime_manifest.json`) reports:

Metric	Full Sample	OOS
Mean IC — Network	+0.005	−0.014
Mean IC — GX	+0.073	+0.041
Mean w_{net}	—	0.249
Range w_{net}	0.000 – 1.000	—

The network signal's OOS IC is negative; the blend accordingly assigns most weight to the GX stream, with w_{net} averaging approximately 0.25.

2.4 Portfolio Construction

Implementation: `08_nalfp/04_portfolio_construction.py`

Each Monday, eligible assets (those with a valid E_{final} score) are ranked and assigned to legs:

Step	Rule
Selection	Top 20% long, bottom 20% short (quintiles)
Position sizing	Inverse 4-week realised volatility within each leg, normalised so long leg sums to +1 and short leg sums to −1 (dollar-neutral)
Single-asset cap	Maximum 5% of gross portfolio per name; excess redistributed pro-rata (iterative, up to 10 rounds)
Cluster cap	Long-leg exposure to any single Louvain cluster 40% of long-leg weight; excess redistributed to under-capped clusters
Turnover budget	Weekly one-sided turnover 30%; smallest $\lvert\Delta w\rvert$ trades are trimmed first (Waterfall mechanism)
Vol target	Gross leverage scaled to target 15% annualised realised volatility, capped at 3× gross

The median portfolio comprises 16 long and 16 short names per week.

2.5 Ablation Variant

An identical portfolio is constructed **without** the cluster-diversification cap (`apply_cluster_cap=False`). This ablation tests whether the network pillar's operational constraint materially affects performance, or whether it serves only as a risk-management overlay.

2.6 Backtest Design

Implementation: `08_nalfp/05_backtest.py`

Parameter	Value
Total sample	48–49 weeks (after 12-week network burn-in)
Training window	24 weeks
OOS evaluation	24 weeks
Cost model	10 bps one-sided per unit of weekly turnover = $0.5 \times \sum \lvert w_{new} - w_{old} \rvert$
Vol annualisation	$\sqrt{52}$ (weekly returns)
Bootstrap	Stationary block bootstrap, block length 4 weeks, 2,000 draws for Sharpe CI

Three benchmark strategies are evaluated alongside NALFP:

1. **EW Momentum Long** — equal-weight long-only top quintile by 4-week momentum
2. **RAAM v2 composite** — IC-weighted composite from stage 04 (if the parquet artifact exists)
3. **plus__lat** — Fama–MacBeth predictions from stage 06 (characteristics + latent controls)

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3. Economic Intuition

3.1 Why Each Factor Should Generate Excess Returns

RC — Crypto Market Factor. The value-weighted return of the entire universe. In the CAPM-analogue setting, its premium *is* the broad-market risk premium. Sign is regime-dependent: positive in bull markets, negative in bear markets. In this sample, RC earned −32.4% annualised (Sharpe −0.78), consistent with a flat-to-bear 52-week period.

SMBC — Crypto Size Factor. Long small-cap, short large-cap. Small-cap names compensate for higher fundamental risk, information asymmetry, and lower liquidity (Fama & French 1993; Hartmann 2025 §3.1). In this sample, SMBC was the strongest performer at +93.7% annualised (Sharpe +3.54), reflecting a pronounced small-cap premium.

MomC — Crypto Momentum. Long recent winners, short recent losers. The strongest documented cross-sectional anomaly in crypto (Liu & Tsyvinski 2022) and equities (Jegadeesh & Titman 1993; Carhart 1997). Investors slowly incorporate information; recent winners continue outperforming. In-sample Sharpe +1.09.

VolC — Low-Volatility (Betting-Against-Beta). Long low-volatility, short high-volatility. Leverage-constrained investors over-bid high-beta assets, depressing their risk-adjusted returns (Frazzini & Pedersen 2014). In this sample, VolC had the highest IC (0.090) and the highest GX cross-sectional t-stat (+3.24) for its risk premium, despite a modest standalone Sharpe of +0.22.

FunC — Fundamental Yield. Long high-fee/high-revenue yield relative to market cap, short low yield. The crypto analogue of earnings yield (E/P). Sparse coverage (~62% missing) limits statistical power; IC = +0.001, GX t-stat = −0.32 (not priced).

SupC — Supply Absorption. Long low-emission tokens, short high-emission tokens. Tokens with low net new supply face less structural sell pressure. Extremely sparse (only 7 weekly observations); excluded from the GX panel and reported for transparency only.

NetMom — Within-Cluster Momentum. Cluster-neutral momentum: long the top half by within-cluster momentum rank, short the bottom half, averaged across Louvain clusters. Isolates idiosyncratic momentum from the broad MomC factor. Sharpe +1.33, GX $\hat{\lambda}_{full} = +0.762\%$/wk (t = +2.42).

NetRel — Cross-Cluster Relative Strength. Long names whose 4-week return exceeds the average of *other* clusters, short names underperforming their cross-cluster peer group. Captures narrative-rotation dynamics (Liu & Tsyvinski 2018). Sharpe +2.24, GX $\hat{\lambda}_{full} = +1.116\%$/wk (t = +2.38). This is the second-strongest factor by risk-adjusted return and the primary channel through which Pillar 1 contributes alpha.

3.2 Why Adaptive Blending Is Theoretically Motivated

The research thesis holds that crypto cross-sectional returns are better predicted by the *interaction* of community structure and factor exposure than by either source alone. When the correlation network is fragmented (high entropy, many small clusters), within-cluster relative performance is the dominant signal — narratives are localised, and idiosyncratic momentum within a thematic cluster is more informative than broad-market factor loadings. When the network converges (low entropy, one or two giant components), common latent factors dominate and the GX expected-return stream should receive more weight.

The IC-proportional blend operationalises this thesis without introducing discretionary parameters: the weight is set entirely by out-of-sample predictive accuracy (rolling IC), and the OOS data determine whether the network stream adds value. As reported in Section 2.3, the network signal's OOS IC is negative (-0.014), causing the blend to tilt heavily toward GX ($\bar{w}_{net} \approx 0.25$). This is an honest empirical finding: in this sample, the fragmentation thesis did not generate positive OOS predictive power.

3.3 Factor Pricing: Observed vs. Latent-Adjusted Premia

The Giglio–Xiu correction is designed to debias observed factor premia when latent factors are omitted. The full-model results show small but instructive changes:

Factor	$\hat{\lambda}^{obs}$ (%/wk)	t(obs)	$\hat{\lambda}^{full}$ (%/wk)	t(full)	$\Delta\hat{\lambda}$
RC	-0.290	-1.56	-0.250	-1.13	$+0.040$
SMBC	$+0.796$	$+3.00$	$+0.591$	$+1.62$	-0.205
MomC	$+1.112$	$+3.44$	$+0.964$	$+1.74$	-0.148
VolC	$+0.985$	$+4.27$	$+0.985$	$+3.24$	-0.000
FunC	$+0.138$	$+0.40$	-0.115	-0.32	-0.253
NetMom	$+0.752$	$+3.37$	$+0.762$	$+2.42$	$+0.009$
NetRel	$+1.079$	$+3.37$	$+1.116$	$+2.38$	$+0.038$

Five of seven observed factors clear the $|t| \geq 1.65$ threshold in the observed-only model: VolC, MomC, NetMom, NetRel, and SMBC. After adding the three latent factors, SMBC's t-stat falls to 1.62 (borderline), while VolC, NetMom, and NetRel remain clearly priced. The network factors (NetMom, NetRel) are essentially unchanged by latent adjustment — their premia are robust to omitted-factor controls, which is consistent with their cluster-neutral construction already accounting for the dominant covariance structure.

The three hidden-factor risk premia ($H1, H2, H3$) are individually insignificant (t-statistics of -0.81, $+0.28$, $+0.13$), with wide 95% confidence intervals. This is expected given the short sample: with $T = 24$ training weeks and 7 observed regressors, the time-series regression has limited power to identify latent factors beyond those already spanned by the observed set.

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4. Assumptions Made

4.1 Theoretical Assumptions

- **Stationarity of factor premia.** The walk-forward GX procedure estimates $\hat{\lambda}$ on an expanding training window and assumes it generalises to the OOS period. The report notes explicitly: "whether the training-window $\hat{\lambda}$ generalises is an empirical question we do not yet have enough data to answer."

- **Spearman IC as the sole regime-detection metric.** The blend weight is determined entirely by rolling 8-week Spearman rank correlation against forward returns. No non-linear regime classifier, macro state variable, or threshold is used. This is a conservative choice but may miss regimes where the signal is non-monotonic.
- **MST + Louvain captures economically meaningful clusters.** The resolution parameter (1.0) and distance metric (Mantegna on Spearman) are fixed. The report identifies 7–11 communities per week but also notes that "we did not observe a clean convergence regime in this slice."
- **Factor zoo exhaustiveness.** Nine factors are specified a priori. The GX procedure accounts for omitted factors via latent PCA, but the Bai–Ng IC_{p2} criterion saturates at the K = 3 cap, suggesting identification is sample-bound rather than data-driven.

4.2 Modeling Assumptions

- **K_{hidden} = 3 is a cap, not a data-driven optimum.** The IC_{p2} information criterion is monotonically decreasing across $K \in \{0, 1, 2, 3\}$; the procedure selects the maximum allowed. Hartmann (2025) used 100+ weeks and selected $K_{hidden} = 7$. Our 24-week training window likely constrains identification.
- **Walk-forward refit frequency = 4 weeks.** The model is re-estimated every fourth week on all accumulated history. This is neither a pure rolling window (which would discard early data) nor a single-shot estimation (which would ignore structural change). The frequency is not optimised.
- **Equal-weight within quintile legs** after inverse-vol weighting. No tilt toward higher-signal assets within each bucket; the ranking is binary (top quintile vs. bottom quintile), not continuous.
- **Cluster cap of 40%** on the long leg is an HHI-equivalent heuristic, not derived from an optimisation framework. The ablation (Section 5) tests its marginal impact.

4.3 Execution and Liquidity Assumptions

- **Weekly Monday rebalance at close.** No intraweek liquidity events, flash crashes, or delayed execution are modelled.
- **10 bps one-sided cost per unit of turnover.** Applied to weekly turnover $= 0.5 \times \sum |w_{new} - w_{old}|$. The report notes this is "intentionally conservative for the size of this strategy (the median name has 30-day ADV > \$1M so 10 bps slippage is fair)" but acknowledges that the long tail of the universe is more expensive in practice.
- **Short-side feasibility.** The strategy shorts the bottom quintile, assuming perpetual/funding markets are available and liquid for

short-side names. No borrow cost or funding rate is modelled.

- **Vol-target scaling** uses the trailing 4-week cross-sectional weighted volatility as a proxy for near-term portfolio risk. Gross leverage is capped at 3×.
- **Remaining data gaps:** Funding rates, perp basis, and exchange-specific flow data are not in the repository. Their absence means the panel relies on Artemis activity, Binance OHLCV, CoinGecko snapshots, and DeFiLlama TVL/stablecoin tables.

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5. Final Results

5.1 NALFP Strategy Performance

Metric	NALFP (full)	NALFP (train)	NALFP (OOS)
Weeks	48	24	24
Annualised Return	+5.9%	+26.5%	−14.8%
Annualised Volatility	11.2%	11.5%	10.3%
Sharpe Ratio	+0.53	+2.30	−1.44
Sharpe 95% CI (bootstrap)	—	—	[−3.45, +1.18]
Maximum Drawdown	−7.1%	−7.1%	−6.6%
Average Weekly Turnover	7.6%	7.8%	7.4%
Hit Rate	60%	67%	54%

5.2 Ablation: NALFP without Cluster Diversification Cap

Metric	No Cluster Cap (OOS)	With Cluster Cap (OOS)
Annualised Return	−17.4%	−14.8%
Sharpe Ratio	−1.69	−1.44
Sharpe 95% CI	[−3.66, +1.11]	[−3.45, +1.18]
Maximum Drawdown	−7.8%	−6.6%
Hit Rate	50%	54%

The cluster cap improves OOS Sharpe by 0.25 and reduces drawdown by 1.2 percentage points. Despite the negative OOS performance, the constraint provides meaningful risk reduction — consistent with the thesis that the network structure is *useful* as an operational portfolio tool even when its cross-sectional signal (within-cluster momentum) does not generate positive IC out of sample.

5.3 Benchmark Comparisons (OOS)

Metric	NALFP	NALFP (no cap)	EW Mom Long	`plus_lat`
Weeks	24	24	25	16
Annualised Return	−14.8%	−17.4%	+58.1%	+129.0%
Annualised Volatility	10.3%	10.3%	43.0%	34.4%
Sharpe Ratio	−1.44	−1.69	+1.35	+3.75
Sharpe 95% CI	[−3.45, +1.18]	[−3.66, +1.11]	[−2.08, +6.61]	[0.51, +8.89]
Maximum Drawdown	−6.6%	−7.8%	−24.8%	−7.4%
Avg. Weekly Turnover	7.4%	7.4%	33.4%	65.1%
Hit Rate	54%	50%	68%	69%

Interpretation: The OOS period was a strongly bullish environment for crypto (EW Momentum Long returned +58.1% annualised). NALFP's vol-target design constrains it to ~10% volatility, which is structurally incapable of capturing the upside of a 58% momentum rally. The plus_lat benchmark achieves a higher Sharpe (3.75) but over only 16 OOS weeks with an extremely wide confidence interval [0.51, 8.89]; this result is not statistically distinguishable from zero at conventional significance levels.

> **Note:** The plus_lat benchmark covers only 16 weeks because it depends on the stage-06 Fama–MacBeth panel which uses a different train/test split (36/16). Direct comparison of NALFP (24 OOS weeks) with plus_lat (16 OOS weeks) should be interpreted cautiously.

5.4 Factor Zoo Performance (Full Sample)

Factor	n (weeks)	Ann. Return	Ann. Vol	Sharpe	NW t-stat	Max DD	IC (char.)
RC	50	−32.4%	41.8%	−0.78	−0.74	−51.6%	—
SMBC	51	+93.7%	26.5%	+3.54	+2.90	−14.9%	+0.015
MomC	48	+40.0%	36.7%	+1.09	+1.16	−24.5%	+0.026
VolC	49	+10.0%	46.2%	+0.22	+0.22	−31.3%	+0.090
FunC	50	+26.6%	31.4%	+0.85	+1.05	−18.4%	+0.001
SupC	7	+51.8%	17.7%	+2.93	—	−1.8%	+0.098
NetMom	40	+30.4%	22.9%	+1.33	+1.19	−19.6%	+0.006

Factor	n (weeks)	Ann. Return	Ann. Vol	Sharpe	NW t-stat	Max DD	IC (char.)
NetRel	40	+74.8%	33.4%	+2.24	+2.14	−18.6%	+0.046

SMBC dominated this sample (small-cap premium), NetRel was the second-strongest factor (narrative rotation), and VolC had the highest cross-sectional IC despite a low standalone Sharpe. RC was negative, consistent with a bearish market factor over the period.

5.5 Stage-06 Econometrics Results (for Context)

The upstream `06_artemis_econometrics` track produced four Fama–MacBeth models with the following OOS performance:

Model	IC Mean	IC IR	LS Mean	LS Sharpe (ann.)	Turnover
`plus_lat`	+0.089	+0.54	+2.65%	+3.74	29.8%
`base_pool`	+0.087	+0.59	+2.87%	+3.46	27.8%
`base_fm`	+0.085	+0.46	+3.35%	+3.90	37.3%
`plus_net`	+0.042	+0.20	+0.70%	+0.68	31.8%

The `plus_net` model (which includes network features in Fama–MacBeth) underperforms the others, consistent with the NALFP finding that network signals do not generate positive OOS IC in this sample. The `plus_lat` model (characteristics + PCA latent controls) achieves the best OOS risk-adjusted return, albeit over only 15 test weeks.

5.6 Stage-07 Hidden Factor Pricing Results (for Context)

Model	Factor	Weekly $\hat{\lambda}$	t-stat	p-value
Fama–MacBeth	`crypto_market`	−0.60%	−0.671	0.502
Latent-adjusted	`crypto_market`	+2.02%	+0.771	0.440
Fama–MacBeth	`crypto_mom`	+1.92%	+1.934	0.053
Latent-adjusted	`crypto_mom`	+1.23%	+0.822	0.411
Fama–MacBeth	`crypto_smb`	+1.91%	+2.189	0.029
Latent-adjusted	`crypto_smb`	+2.32%	+1.970	0.049
Fama–MacBeth	`crypto_tvl_orth`	+0.19%	+0.087	0.931
Latent-adjusted	`crypto_tvl_orth`	+1.55%	+0.735	0.463

Only `crypto_smb` survives the latent adjustment (sign preserved, $|t| \geq 1.65$). This is a more conservative test than NALFP's factor-specific walk-forward pricing, and it uses a different factor construction methodology; the two results are not directly comparable but provide complementary evidence.

6. Further Potential Research

6.1 Sample Size and Robustness

- **Extend the data window.** 52 weeks (24 OOS after burn-in) is insufficient for reliable factor-model estimation. The Sharpe-ratio standard error on 24 weeks is approximately $1/\sqrt{24} \approx 0.20$; the bootstrap CI for NALFP OOS Sharpe is wider still and crosses zero. Extending to 2–3 years of data would stabilise $\hat{\lambda}$ estimates and provide more power for K_hidden identification.
- **Rolling walk-forward.** Replace the single 24/24 train/test split with a proper expanding or rolling walk-forward, generating many more OOS data points for both IC and Sharpe evaluation.

6.2 Signal Construction

- **Non-linear regime detection.** The current blend uses linear IC-weighting. A gradient-boosted classifier, regime-switching model, or conditional copula could capture non-linear interactions between network fragmentation and factor exposure.
- **Alternative network construction.** The MST + Louvain pipeline uses Spearman correlation with fixed 12-week window and resolution = 1.0. Alternatives include: (i) Granger-causality or transfer-entropy networks, (ii) dynamic time-warping for similarity, (iii) adaptive resolution or hierarchical clustering. The stage-06 report explicitly suggests "replace the rolling-correlation cluster with a Granger-style predictive link network."
- **Stablecoin inflow as a regime indicator.** The `stable_inflow_z` feature is computed in stage-06 and referenced in the NALFP `_common.py` `REGIME_COLS`, but it is not used in the final blend. Incorporating it as a macro-state variable could improve regime detection.

6.3 Portfolio Construction

- **Continuous position sizing.** Replace the binary quintile cut with position sizes proportional to expected return (e.g., conditional mean-variance or risk-parity on signal strength), which would extract more information from the ranking.
- **Per-asset transaction cost modelling.** The flat 10 bps cost is a rough approximation. Modelling per-name slippage as a function of 30-day ADV, bid–ask spread, and funding rate would more accurately penalise illiquid short-side names.

- **Short-side feasibility.** Explicitly modelling borrow availability, funding rates, and perp basis would constrain the short leg to assets where shorting is actually implementable.
- **Alternative cluster cap.** The 40% cluster cap is heuristic. An optimisation framework (e.g., minimum-variance with cluster-aware covariance shrinkage) could set this endogenously.

6.4 Factor Model Improvements

- **IPCA for time-varying loadings.** The v1 approach (Kelly–Pruitt–Su 2019) was abandoned for interpretability but could be layered on top of GX for *beta dynamics* even if pricing remains GX. Time-varying loadings may capture regime-dependent factor exposures that static betas miss.
- **Factor timing.** Dynamic $\hat{\lambda}$ estimation conditioned on macro-state variables (e.g., stablecoin flows, BTC dominance, DeFi TVL growth) rather than static expanding-window averages.
- **Additional factors.** The current zoo omits on-chain activity growth (DAU growth) as a standalone factor; the RAAM v2 `G` factor had a negative IC (-0.04) in the factor scoreboard but a reformulated version using per-protocol DAU growth relative to market cap could be more predictive. A funding-rate/perp-basis factor would capture the cost-of-carry dimension entirely absent from the current specification.

6.5 Risk Management

- **Downside risk controls.** The 15% vol target caps volatility ex ante but does not incorporate tail-risk measures (CVaR, max drawdown limit, or circuit-breaker rules). Adding a conditional volatility model (GARCH or EWMA) could improve vol targeting in clustered-risk episodes.
- **Correlation stress testing.** The portfolio's diversification benefit assumes cluster structure is stable. Stress-testing under scenarios of correlation breakdown (e.g., all clusters merging in a market crash) would quantify the tail risk of the cluster cap.
- **Leverage and margin constraints.** Perpetual futures venues impose maintenance margin and auto-deleveraging. The 3× gross leverage cap models this loosely; a more detailed margin simulation would improve implementation realism.

Appendix A: Items Requiring Confirmation

The following items could not be fully confirmed from the codebase alone and require further documentation or manual review:

1. **RAAM v2 composite IC weights.** The `04_factors/README.md` reports pre-fix values (V=+0.21, C=+0.18, S=+0.12, M=+0.06, F=+0.02, T=+0.00, G=−0.04), but notes "the table reflects results before the A1/A2 correctness fixes." The current values after re-running may differ and should be confirmed from the regenerated figures.
2. **BTC direction model integration.** `05_btc_direction/` is a standalone supervised classifier not integrated into NALFP. Whether it should be discussed as a separate track or referenced as a potential regime indicator for future work requires clarification.
3. **Stage-06 `plus_lat` model specification.** The backtest references `plus_lat` as a benchmark with Sharpe 3.75 on 16 OOS weeks — this is the same model from `06_artemis_econometrics/05_models.py`. The exact specification (which characteristics, which latent controls) should be confirmed from that stage's artefacts.
4. **Exact OOS date boundaries.** The manifests record 48–49 total weeks and 24-week training, but the precise start/end dates for the OOS evaluation window are not explicitly stated in the manifest JSONs.
5. **Competition deliverables alignment.** The competition requires three separate submissions: (a) a Research Report in PDF format, (b) a GitHub repository with reproducible code, and (c) a Pitch Deck (Google Slides) summarising the strategy for a non-technical audience. This document serves as the Research Report; the code repository and pitch deck are separate deliverables.

Appendix B: Reproducibility

The full pipeline can be reproduced by running:

```
python3 08_nalfp/01_network_dynamics.py
python3 08_nalfp/02_factor_pricing.py
python3 08_nalfp/03_regime_detector.py
python3 08_nalfp/04_portfolio_construction.py
python3 08_nalfp/05_backtest.py
python3 08_nalfp/06_report.py
```

All artefacts are written to `08_nalfp/artifacts/data/` (parquet + CSV), manifests to `08_nalfp/artifacts/manifests/` (JSON), and figures to `08_nalfp/figures/` (Plotly HTML, with PNG saved when `ARTEMIS_SAVE_PNG=1`). The upstream stages (`06_artemis_econometrics/`, `07_hidden_factor_pricing/`, `04_factors/`) must be run first to produce the required input parquets.

Appendix C: Honest Limitations and Critical Evaluation

The Artemis competition emphasises that *"a strategy with a negative Sharpe ratio that is correctly understood, honestly evaluated, and clearly presented will score better than a strategy with impressive backtested returns and no critical analysis."* In this spirit, we catalogue the strategy's failures and fragilities alongside its strengths.

What worked

- **Factor identification is statistically meaningful.** Five of seven observed factors are priced at the $|t| \geq 1.65$ threshold in the observed-only Fama–MacBeth cross-section; four survive latent-factor adjustment (VolC, MomC, NetMom, NetRel). The pricing framework is not pattern-fitting — it is grounded in published economics (size, momentum, low-vol, network structure).
- **The network pillar is operationally useful.** Even though the network signal's OOS IC is negative, the cluster diversification cap demonstrably improves OOS risk metrics (Sharpe improves from −1.69 to −1.44; drawdown improves from −7.8% to −6.6%).
- **The strategy's vol target works as designed.** NALFP delivers single-digit volatility (10.3% annualised OOS), which is a legitimate risk-management outcome even when returns are negative. For risk-averse allocators, the max drawdown of −6.6% is modest relative to benchmarks (EW Momentum Long: −24.8%).

What failed

- **OOS returns are negative.** NALFP's OOS Sharpe of −1.44 (95% CI: [−3.45, +1.18]) means we cannot reject the null that the strategy has zero risk-adjusted return on this sample. The bootstrapped confidence interval spans zero.
- **The network signal does not predict OOS.** Within-cluster momentum (the primary Pillar 1 cross-sectional signal) has an OOS IC of −0.014. The adaptive blend reduces w_{net} to ~0.25, effectively weighting the strategy toward the GX stream, but even the GX stream's OOS IC of +0.041 is modest.
- **Factor premia reverse out of sample.** The training window (weeks 1–24) had a strong size premium (SMBC Sharpe +3.54) and negative market factor (RC Sharpe −0.78). The OOS window appears to feature a different regime — one where the long/small-cap tilt inherent in SMBC and the short/market tilt in RC both contribute negatively. This is precisely the regime-shift risk identified in the research plan.

Structural limitations

- **Sample size.** 52 weeks total → 24 OOS weeks after the network burn-in. The bootstrap CI for NALFP OOS Sharpe crosses zero. The SE(Sharpe) $\approx 1/\sqrt{24} \approx 0.20$; detecting a Sharpe of 1.0 with 80% power would require ~64 OOS weeks at this volatility.
- **Hidden factor identification.** Bai–Ng IC_{p2} saturates the $K = 3$ cap, suggesting the $T = 24$ training window is short for identifying hidden risk premia. Hartmann (2025) used 100+ weeks and selected $K_{hidden} = 7$.
- **TVLC and SupC.** Excluded from GX due to coverage; their stats in the zoo table are computed on 0 and 7 weeks respectively and should be read with extreme caution.
- **Costs.** 10 bps one-sided on turnover is reasonable for large-caps; the long tail of the universe is more expensive in practice.
- **What would break this.** A regime shift that flips the sign of SMBC or MomC out-of-sample (precisely what is observed in some OOS weeks). The strategy's strength is the *factor structure identification*; whether the training-window $\widehat{\lambda}$ generalises is an empirical question that 24 OOS weeks cannot definitively answer. Concretely: in a bullish low-volatility regime where small-caps dominate, NALFP is likely to underperform a simple long-only momentum strategy because its dollar-neutral, vol-target structure caps both participation in the rally and the magnitude of drawdowns.